

Digital Access, Digital Literacy, and Afterlife Preparedness: Societal Contexts of Digital Afterlife Traces

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Abstract

This study aims to evaluate how individuals are prepared to cope with or plan for their afterlife digital footprints, by examining how (1) access, (2) literacy, and (3) preparedness for digital afterlife work in concert to influence one's wellbeing. We found the indirect relationship between access and wellbeing and the influence of digital literacy on wellbeing was indirect, illustrating that the key to the puzzle is in the sequential step in which digital literacy incubates the readiness to cope with digital remains, which influences one's subjective wellbeing. One of the most unrecognized challenges facing digital traces is the exploitation of post-life remain of data, as the question of who accesses, owns, or controls personal data after death remains largely unanswered. We argue that the preparedness for digital afterlife represents a new form of social concern with real-life consequences, or even a newer space for inequality debates.

Keywords

digital afterlife, digital inequality, capacity building

Individuals are increasingly assuming a responsibility for coping with all facets of their digital lives. Protecting perpetual traces of digital footprint and privacy, in particular, remains subject to individual action under service terms and agreements arranged by social media platforms. Under the U.S. self-regulation, digital platforms, such as Facebook, Google, and Instagram, regulate personal data on their own terms and can exploit them even after a person stops using services. That is not to say that individuals do not have any agency or the means of control (Bassett, 2022; Harbinja, 2022). However, as our digital lives are in the hands of few unregulated platforms and their increasingly algorithm-driven environments amplify the risk of losing our abilities to define self/selves—not just while we actively use their services but also after our uses stop or even when our lives come to an end. Fundamental is the incongruence between a person's biological and digital lives as the ontology of digital traces defies the finite presence of humans as a biological being (Morse, 2018). Put differently, we are mostly by de facto left alone in our lifetimes to respond to unprecedented challenges to digital identities, privacy, and associated assets and to delineate the scope of our digital lives in the future.

Our goal in this article is to evaluate digital afterlife preparedness—defined as how individuals are prepared to plan, manage, or cope with the afterlife of their digital footprint and its remains about themselves (Morse, 2018; Morse, 2024;

Morse & Birnhack, 2022). Specifically, we investigate (1) whether one's levels of digital access and digital literacy are associated with better planning for digital remains and (2) whether such digital preparedness is related to one's perceived wellbeing. Our contribution is to look at these issues from the perspective of digital inequalities (Chen et al., 2014; Zillien & Hargittai, 2009) and analyze the formation of digital afterlife readiness as an individuals' digital capacity-building process. Digital afterlife, the remains of personal data after death, generated much attention (Bassett, 2022; Birnhack & Morse, 2022; Harbinja, 2022). Understanding this issue from the perspectives of ordinary users, however, has been still relatively scarce, especially from the lens of digital inequality.

Central to our contribution is the conceptualization of unequal distributions of digital and non-digital wellbeing in terms of third-level digital divide (van Deursen & Helsper,

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2015). That is to say, there is a broad range of social disparities perpetuated by individuals' digital capacities, with favorable or unfavorable outcomes resulting from the differences in the levels of digital access, digital literacy, and other areas of digital resources or capitals—digital afterlife preparedness in our study. We aim to show how digital access, digital literacy, and digital afterlife preparedness are interlaced in sequential and compound ways. In so much as digital afterlife preparedness might be an important predictor of tangible life outcomes, such as one's wellbeing, we can identify it as one of the key factors explaining the reinforcement of inequalities, thus paving the ways to understanding the process of deepening social inequalities (1) between online/digital and non-digital resources and (2) between one realm of digital life and another.

In what immediately follows next, we provide a brief background of digital afterlife industry practices to contextualize this study's empirical analysis. Then, we assess the extent of digital afterlife preparedness among U.S. populations and identify key components of building individual capacity.

Background: Digital Traces and Commercial Platform

The complexities of digital footprints and remains lie in the fact that personal data outlive the data subject (Morse, 2018; Öhman & Floridi, 2018; Park, 2021a). Permanent digital traces pose numerous challenges, such as the evolution of digital industry exploiting remnants of personal data and associated information. Some U.S. states have enacted the Fiduciary Access to Digital Assets Act, which regulates access to digital remains. This is a remarkable step forward. As of today, however, regulatory environment that governs the commercial use of digital traces, their privacy, and derivative databases after death remains limited because protection laws center upon assets, not information aspects, of data (Park et al., 2020). Furthermore, although some platforms such as Apple, begun Legacy Contact options, the approach has been reactive, that is, putting individual users in charge in the case of post-life problems, as opposed to being proactive, that is, putting regulatory prohibition on the use of data.

Harbinja (2022) raised this concern about privacy of online identity, data protection, and access to data after death and suggested there is a societal need to put together stronger regulatory protection as much as practical digital tools. The relatively weak regulatory protection for post-life privacy as in U.S. regulatory environment, for instance, means that users, in their use of digital platforms, often fight on their own against the power of commercial platforms. In other words, users have little say in controlling the post-life digital traces that affect their reputation, personal integrity, and privacy after death. In this context, Birnhack and Morse (2022) argued for the notion of posthumous privacy (privacy after life) and for having reasonable privacy expectations of the living people regarding their post-mortem condition.

It is no surprise that the question of who accesses, owns, or controls data after death is still largely contested. In fact, the birth of afterlife digital industry is symptomatic of these challenges unforeseen in the previous era of Web 1.0 (Bassett, 2022; Kneese, 2023). Online services sprang up for those who worry about post-life digital traces as people want to deposit online passwords or digital assets such as bitcoins safely in case of a sudden death or medical emergency. Data firms also popped up to help deal with digital remains after death, with some services available for a user to arrange specified messages to be sent to their selected recipients posthumously (Evans, 2018; Morse, 2018). These are some examples of commercial spaces that are specifically carved out for high-end niche demands at time of a rare, sad, and emotionally-drained event like a person's death.

Nevertheless, our immediate concern in this article is on the mundane contexts in which ordinary people use social media, such as Facebook, Twitter, or Instagram, for their routine activities—for instance, liking, sharing, as much as posting their comments and pictures, joining discussion groups, chatting via direct messaging, and so on. This extends to user's right to data, for instance, when Facebook terminates or suspends their use of a digital service or even when users deactivate their social media accounts willfully. We have several reasons.

First, despite a growing presence of afterlife industry, most users do not use those special services, whereby only few people enjoy privileged services. Instead, people's digital presence (including their post-life traces) is confined in a handful of popular platforms, such as Google or Facebook, where the vast majority of digital participation is concentrated. Second, digital ecosystem, interconnected in very few highly trafficked platforms and proprietary algorithms, raises the issue of how vulnerable data have become to a whim of commercial surveillance. According to Bassett (2022), this amounts to "the second loss" of self/selves due to the lack of a sense of personal autonomy over digital traces. Fundamentally, we are concerned about the transition into a "big data" algorithm era in which data collection already became part of our ontology even after death—but confronted by the failure in recognizing social aspects of person's afterlife (Harju & Huhtamäki, 2023; Kneese, 2023).

Digital Afterlife Preparedness: Individual Users in Commercial Platforms

Here, the distinction between institutional and interpersonal privacy is important (Lutz et al., 2018). The privacy of digital afterlife is often reduced to be an interpersonal matter for families or friends trying to gain access to the lost one's digital records posthumously. Although interpersonal traces are significant pieces of a person's afterlife identity, users do face far more complicated realities in institutional environments where ambiguous corporate service terms dictate the user data condition—namely, how metadata, such as behavioral traces, will be retained and used for what purposes. We push

this discussion further into the direction of investigating individual capacities in actual consumption. Scholars (e.g., Harju & Huhtamäki, 2023; Kasket, 2020) noted that data are not necessarily immortal given the diverse dimensions of personal data associated with one's emotion that is also to cease at some point. The issue to many ordinary users, however, is that personal data persist to live beyond their lifespan. The growing population of older social media users, for instance, produces an increasing need for assistance with key decisions with regard to their digital traces. In fact, the potential consequence of a poor capacity related to managing one's digital traces can be serious beyond a violation of a person's privacy (Harbinja, 2022; Morse & Birnhack, 2022).

Sociological literature (e.g., Hockey, 1990) looked at the societal process of bereavement and grief and how the meaning of loss evolves among surviving families, friends, or relatives. Subsequently, one area of inquiry has been coping (as much as maintaining their wellbeing) strategies in their effort to continue symbolic relationships with those who passed away. Scholars also paid attention to the development of social norm about how to define and manage consequences of death (Harju & Huhtamäki, 2023). A classical Durkheimian analysis by Warner (1959), for example, provided anthropological insights into the end-of-life practices at hospitals, as his work identified unique communicative patterns among people facing death. The lesson of his study is that the acquisition of coping skills, which reflected the traits in particular environments, is fundamentally societal and, thus, would have never developed if people lived in isolation as separate individuals.

Recent studies strike similar chords. Gibbs et al. (2015), for example, examined photo-sharing coping practices in Instagram, echoing broader shifts heralded by social media into personalized informal practices. A study by Savin-Baden (2021) recognized the role of digital platform—as societal environment—in changing natures of afterlife, its understandings and practices, prompting to consider perpetuity of being human in algorithmically transformed forms. This line of insight carries significance to this study's purpose. As most social activities migrated to digital platforms, the acquisition of trait, skill, or value anchored in one's socialization will be even more strongly tied to digital activities. Extending this logic, we argue that the debate about digital afterlife should move from a somewhat clinical or remedial account of interpersonal coping strategies among close friend-family members. Instead, we propose that the decisions regarding informational remains, as much as human biological remains, should not be regarded merely as an episodic transactional matter in marketplaces but as a capacity-building process constitutive of unique social resources that harness individual abilities, with corresponding consequences (Bourdieu, 1984). In other words, individual capacity is built based on a socializing process as the nature of the social support system and resources helps shape the decision to prepare, plan, or cope with the afterlife of digital footprints.

Conceptual Focus: Digital Access, Digital Literacy, and Afterlife Preparedness in Cumulative (Dis)Advantages

We build a conceptual lens based on the perspectives of digital inequality. The distinctions between the first-, second-, and third-level divides provide a precise understanding of how complex dynamics of social conditions function to shape the capacities related to digital remains from the standpoint of a user. We can start with the differences in (1) infrastructural access (first divide) and (2) digital skills and usage patterns (second divide; Zillien & Hargittai, 2009). The idea is that the disparities, as manifested in the first and second levels, are not randomly produced but are the outcomes of systematic disadvantages that perpetuate certain social groups (van Deursen & Helsper, 2015). The third-level divide, reminiscent of the preceding two divides, posits that there will be persistent differences in translating digital advantages to life objectives in various areas of concern, such as afterlife preparedness, even after certain levels of access to digital infrastructure and digital skill sets are achieved.

Here, the two processes of (1) sequential and (2) compound digital exclusion help understand the mechanism in which respective levels of digital divides collectively reproduce inequalities. First, sequential exclusion explains how different types of digital divide depend on one another, with respective disadvantages sequentially moving from one level to another. Compound exclusion, on the other hand, can be understood as the process of accumulating multifaceted disadvantages on the same level of divide, when the divide in one type also lacks closely-related other types of divides. Surely, the mechanism, in which the divides in various domains of digital and non-digital life outcomes are jointly connected and eventually produced, might differ by context or issue. However, this insight of sequential and compound exclusion is illuminating because identifying different processes of divides parcels out otherwise conflated steps that incubate digital inequalities, and this way, we can identify precisely where potential intervention is necessary.

In the spirit of preceding discussion, we advance a set of hypotheses related to individual capacities to plan for, or cope with, the afterlife presence of digital footprint and its remains. Figure 1 summarizes our conceptual model hypothesizing the first, second, and third divides as sequential and compound processes are closely interlinked to each other. The first link in our model hypothesizes (1) the influences of demographic and socioeconomic backgrounds on digital access and (2) the positive role of digital access in the formation of afterlife preparedness. This first set of hypotheses seeks to verify the role of digital resources at the infrastructure level and unequal socioeconomic-demographic bases of access. A Pew report (Anderson & Kumar, 2019) suggested that there are still widespread online access disparities

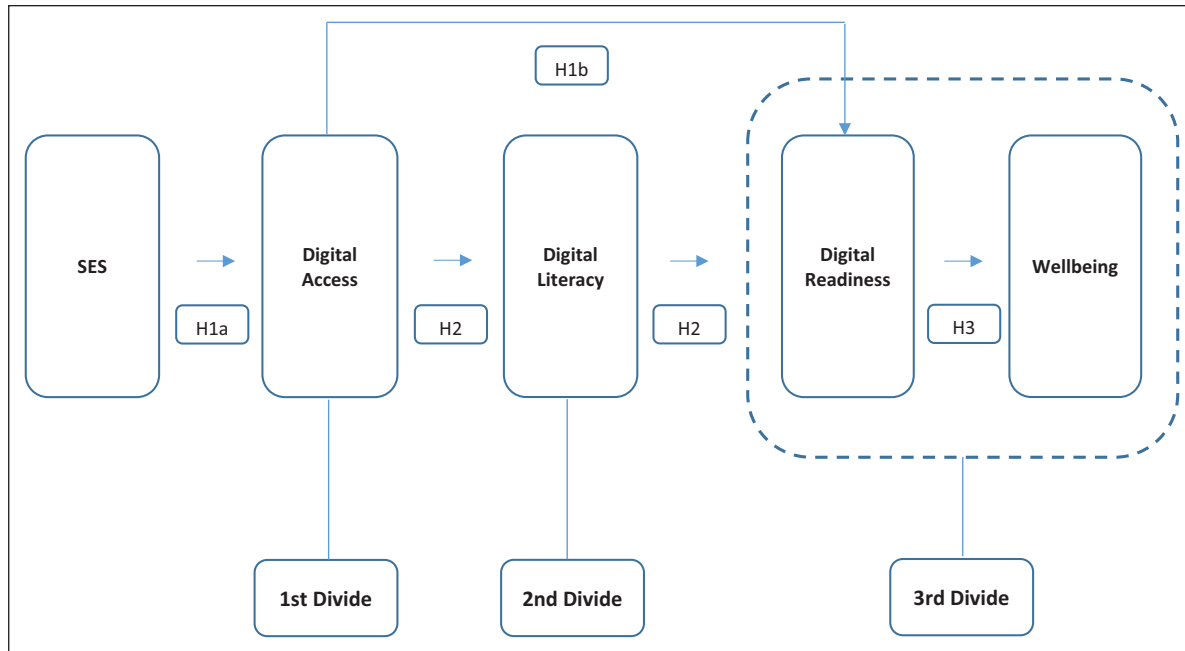


Figure 1. Hypothesized formation of digital readiness and its consequence.

between those “haves” and “have-nots” in the United States, which perpetuates digital disadvantages in other domains, such as job and health information search. Accordingly, we predict that there will be a direct influence from the level of access on individual readiness for digital remains, as those surrounded by better socioeconomic-demographic resources, indicated by the five frequently used indicators such as education, income, race, gender, and age (Van Deursen et al., 2017), might enjoy better access, and thus the better infrastructural bases to build individual capacity.

Hypothesis 1a (H1a). A higher level of digital access will be positively associated with a higher level of readiness for digital remains.

Hypothesis 1b (H1b). However, there will be persistent demographic and socioeconomic divides in accessing the infrastructural basis to build readiness as those less educated, less wealthy, women, non-Whites, or elderly will be associated with a lower level of digital access.

The second link in our model posits a connection between digital access and digital literacy, which in turn will help incubate the abilities to prepare for their digital remains. This suggests that the influence of digital access is indirect through the second-level divide (digital literacy), which will play a critical role in translating one’s access to infrastructural resources into higher capacities of afterlife preparedness. Studies (Livingstone, 2011) consistently discovered the role of having better access in producing the differences at the level of digital literacy. In addition, a higher level of digital literacy has been found to be a significant predictor of

better digital skillsets in various domains, such as internet privacy management (Wendt et al., 2023). Applying this to the context of afterlife preparedness, we can predict that those with a higher level of digital access will enjoy a higher level of digital literacy skill sets based on which to build the better capacity in dealing with digital remains.

Hypothesis 2 (H2). A higher level of digital access will have a positive, indirect influence on the individual readiness for digital remains through digital literacy.

The third link in our proposed model connects the level of digital readiness to the non-digital life outcome, assessing the influence of third-level divide (readiness) on perceived wellbeing. Studies (e.g., Ahn & Shin, 2013) began to discover that digital inequalities, indicated by various measures such as skills and digital connectivity, have real consequences in everyday life. Notably, a study by Büchi et al. (2018) found that a subjective quality of life depends on individual perception of digital belongingness (whether a person sees that she/he belongs to information society in general). Extending this, it is possible to see how the perceived ability to cope with digital remains might help predict one’s own assessment of quality of life, as people see themselves in a position to be digital ready, digitally engaged, or even fully included. Fundamentally, this opens up a possibility that individual readiness or self-evaluation about one’s capacity acquired from digital literacy can bring about tangible effects on life outcomes.

In assessing influences from the readiness for digital remains, we make a distinction between general and privacy-specific readiness. The process of building individual

readiness might be compounded as the two distinct but related types of readiness have separate routes of sequential influence, while relaying the function of digital literacy differently. On the general level, readiness might be a product of how individuals perceive their overall readiness, taking into consideration all relevant matters in their online and social media skills, habits, and access to several digital platforms. On the specific level, privacy might challenge users with its unique set of issues that can be distilled from specific protective need, and it can potentially exclude those who do not see themselves prepared from developing a sense of wellbeing. Privacy involves complex decision-making regarding personal data preferences, with their distinctively paradoxical effects (Neubaum et al., 2023). Thus, in assessing the influence of individual readiness, a careful dimensional distinction will be necessary so that we do not conflate or compound two distinctive capacities that have different perceptual consequences.

Hypothesis 3a (H3a). A higher level of readiness resulting from digital literacy will have a positive influence on a non-digital life outcome indicated by perceived wellbeing.

Hypothesis 3b (H3b). However, there will be compounded readiness (general as opposed to privacy-specific) through which different patterns of indirect influence of digital literacy occur.

Finally, in assessing the link from readiness to perceived wellbeing, we turn to the possibility that the proposed relationship might be stronger (or weaker) depending on two factors: (1) age and (2) privacy-worry. First, van Deursen and Helsper (2015) documented that older people are often left at a disadvantage on key dimensions of life as they are prone to health problems and less disposable income, and those characteristics get transferred to their online engagement with cumulative negative effects. Older users (Elueze & Quan-Haase, 2018) may forecast (and have a reason to care about) pitfalls from digital remains more immediately than younger people, with the level of their readiness likely to affect perceived wellbeing more strongly. Second, it is plausible to expect that the level of privacy-related worry affects perceived consequences of being ready for digital remains, with the ability to cope with (thus control) data influencing people with higher privacy-worry more strongly than those who are less worried. As in the case of age, we also see the potential interaction in which the function of readiness for digital remains in perceived wellbeing will be moderated by one's negative appraisal—how much a person is worried about digital footprints in her or his use of digital devices and social media. That is to say, there might be potentially magnifying interactive influence of two factors.

Hypothesis 4 (H4). The positive influence of individual readiness on perceived wellbeing will be greater for (1) older users and (2) those who are worried about the privacy of digital remains.

Methods

The data for this study were collected in the United States, September 2018, and the questionnaire was administered through Qualtrics on its online platform. The survey had two attention-check questions, and we included only the respondents who passed those questions for analysis. Valid responses ($n = 824$ adults) were identified out of 978 completed responses. Compared to the U.S. census, our sample had fewer non-Whites, 2013–2017 American Community Survey (ACS) = 27%, this study = 19.2%. Median household income was between \$40,000 and \$50,000 (5 on a scale of 12), which was a little lower than ACS = \$57,652. Nevertheless, overall demographic characteristics were similar to those of the U.S. population and surveys administered by other organizations (e.g., Pew internet & American Life Project, 2018). Percentage of females was 51.9% (ACS = 50.8%). Age varied between 18 and 87 years ($M = 46.74$, $SD = 16.47$; ACS = 45–54 years), with median educational attainment for some college (3 on a scale of 6; ACS = some college).

Measures: Digital and Non-Digital Wellbeing

Our analyses included five groups of predictors illustrated in our model, registering the distinction between digital wellbeing (a. digital access; b. digital literacy; c. readiness for digital remains; d. digital privacy-worry) and non-digital wellbeing (e. perceived quality of life or wellbeing, with socio-demographic backgrounds). All measures came from previously-validated items anchored with conceptual underpinning (access, Hargittai & Hinnant, 2008; literacy, Hargittai, 2022; privacy-worry, Park & Jones-Jang, 2023; wellbeing, Diener et al., 1985). We developed readiness for digital remains by modifying a general measure of digital readiness (Van De Werfhorst et al., 2022) into this study's context. While this item development did not include focus group/interview data, our testing indicates predictive validity. Surely, this item will be needed to get further validated with future studies.

Digital Access. In operationalizing digital access, we modified a measure of autonomy (Hargittai & Hinnant, 2008). The idea is that the greater the number of access points, often indicated by the number of different locations where a person can gain online access, the greater the degree of individual autonomy and, potentially, agency the person exercises. The actual question was: "At which of these locations do you have access to the internet, that is, if you wanted to where could you use the internet?" We provided the respondents with nine options: library or computer lab, school or campus, workplace, home, friend's home, family member's home, coffee house/internet café, community center, and on the go (using cell phone, smartphone, tablet, iPad, or roaming wireless). We counted a total number of different locations that each respondent reported and used that number as digital access ($M = 3.48$, $SD = 2.23$, min. = 1, max. = 9).

Digital Literacy. For the measure of digital literacy, we used 17 digital terms, and respondents were asked to rate the level of familiarity with each of them (Hargittai & Hsieh, 2012). Understandings about each term were assessed on a five-point scale (1 = *no understanding* to 5 = *full understanding*). Those terms were preference settings, meme, tagging, privacy settings, viral, followers, hashtag, advanced search, PDF, spyware, wiki, JPG, cache, malware, and phishing. Scores for all items were combined to calculate the total ($M = 49.09$, $SD = 16.96$, min. = 14, max. = 75, $\alpha = .95$).

Digital Readiness. Being a critical part of our analysis, this variable aimed to register a level of individual preparedness for digital remains. As hypothesized earlier, we measured the potentially compound dimensions in two ways: (1) general preparedness and (2) privacy preparedness. As mentioned earlier, these measures were newly developed for the purpose of this study by adopting prior studies' digital confidence measures (Hargittai & Hsieh, 2012; Van De Werfhorst et al., 2022).

First, general preparedness, measured in a five-point scale item (1 = *not at all*, 5 = *very confident*), asked the respondents to rate the extent of confidence in dealing with digital remains in their use of each of the following: (1) social media account, (2) mobile/smartphone, (3) text messages, (4) emails, (5) personal computer, (6) tablet, and (7) smart speaker, such as Google Home, Amazon Alexa, or Echo. All responses were combined to generate a single index (general preparedness: $M = 23.40$, $SD = 6.86$, min. = 5, max. = 35, $\alpha = .95$). We provided respondents with the general description of digital remains before asking how they feel they are ready. The wording was: In general, please indicate the extent to which you feel prepared for protecting digital remains in your use of each of the following items (seven items were offered). Second, privacy preparedness was measured on a five-point *single-item* scale (1 = *not at all*, 5 = *very confident*), specifically centered on the privacy dimension asking the respondents to rate the extent of their confidence in protecting the privacy of digital remains ($M = 3.48$, $SD = 1.01$, min. = 1, max. = 5). The wording was: How confident are you in protecting your privacy of digital remains or those traces, such as pictures, videos, email, tweets, or texting messages?

Digital Privacy-Worry. The measure of privacy concern (Park, 2022; Park & Jones-Jang, 2023) was modified in the context of digital remains. A brief scenario concerning the potential violation of privacy and unauthorized access to digital footprints after death was described. Then, the respondents were asked to report the presence or absence of their worries in their respective use of each of digital devices or services: email, text, social media, mobile/smartphone, and so on. We coded Yes (worried) into 1, with rest of the options (not worried, no such use, or do not know) as 0, and then, we created an index measure by summing up all scores ($M = 1.84$, $SD = 2.42$, min. = 0, max. = 7, $\alpha = .92$).

Perceived Wellbeing and Socio-Demographic Backgrounds. We measured the wellbeing as the final outcome variable. A five-point scale was used, ranging from 1 (*do not agree at all*) to 5 (*completely agree*), with five items from the Satisfaction With Life Scale (Diener et al., 1985). We combined all scores into an index measure ($M = 16.85$, $SD = 4.63$, min. = 5, max. = 25, $\alpha = .895$). Five variables about respondents' demographic (age, gender, and race) and socioeconomic (income and education) backgrounds were used. Their distributions were reported in the sample description provided earlier.

Analytical Strategies

We first break down the distribution of digital wellbeing (access, literacy, readiness, and privacy-worry) by non-digital (demographic and socioeconomic) attributes to highlight the baseline statuses, and this was followed by the mean tests performed to detect subgroup differences. To test the hypotheses, a series of step-wise OLS (Ordinary Least Square) regressions were run, with a group of predictive variables added in sequential steps. These steps followed the order of hypotheses (H1 through H3), starting from (1) access, (2) digital literacy, and (3) digital readiness. In addition, for H4, we created four interaction terms between each of the two compound dimensions (general and privacy preparedness), privacy-related worry and age, respectively.

Statistically speaking, sequential steps proposed in our model presuppose indirect relationships, but we are also guided by the need to observe precise patterns in which addition of variables in each step suppresses or inflates variances in distinct regressions. Interaction allows us to inspect how influence from one variable depends on the other variable, as in a 2 (old/young) $\times$ 2 (low/high) composite. Thus, first using step-wise regressions, we inspected each step separately and then employed PROCESS to highlight a summary view of the proposed model. The goal is to understand the process involving the formation and consequences of readiness for digital remains. This is a parsimonious way to answer how readiness in a specific domain is formed in a succession of unequal distribution of digital resources, as they are rooted in non-digital demographic-socioeconomic factors, resulting in tangible real-world consequences.

Results

Table 1 presents descriptive summaries. These are the mean scores reported of combined index measures, also divided by each demographic characteristics. The level of general preparedness remained relatively modest ($M = 23.40$, min. = 5, max. = 35), with less than a half of the respondents (43%) reporting their readiness in coping with digital remains. Privacy preparedness was modestly present ($M = 3.48$, out of 5), but this indicates that just above 34% of respondents reported their confidence in dealing with the potential privacy violation of digital remains. These results are slightly different from other works that found higher awareness in

Table 1. Digital Wellbeing Status by Non-Digital Sociodemographic Status.

Digital wellbeing indicator	Full sample	Age older than ≥ 46.74	Women	Non-White	Household income less than \$50,000	Some college or less
Digital readiness						
General ready (5–35)	23.40	23.02	23.37	22.56	22.22***	22.61***
Privacy ready (1–5)	3.48	3.35***	3.47	3.39	3.35**	3.40**
Digital privacy-worry (0–7)	1.84	1.40***	1.59**	2.30**	1.64*	1.68*
Digital literacy (14–75)	49.09	44.11***	47.45**	49.41	45.49***	46.44***
Digital access (1–9)	3.48	3.08***	3.48	3.03**	2.97***	3.16***
	<i>N</i> = 824	<i>N</i> = 404	<i>N</i> = 428	<i>N</i> = 158	<i>N</i> = 368	<i>N</i> = 522

Note. Mean test results between subgroups are shown.

* $p < .05$, ** $p < .01$, *** $p < .001$.

this matter, probably indicating that the U.S. population in general (especially at the time of this survey's data collection) was not aware of this issue as those in other nations (Morse, 2018, 2024). This will warrant survey works, such as cross-cultural differences, in future. Compared to general and privacy preparedness, the level of privacy-related worry was low as well ($M = 1.84$, out of 7), indicating that there is a low level of public awareness regarding the matter of digital remains. Only 30% of respondents said that they worried about the digital traces they leave behind in their use of social media, and as little as 12% of them reported their worries in all seven forms of digital remains.

Mean tests in Table 1 showed that even the low level of digital readiness turned out to be available only among those in better socioeconomic positions. Group differences between educational and income levels were huge and consistent, with those with lower income and some college education (or less) falling behind all indicators of readiness, privacy-worry, access, and digital literacy. Non-Whites reported higher privacy-worry ($M = 2.30$, compared to $M = 1.84$ in the total sample), but they lagged behind in digital access—the most basic infrastructural basis based on which individual capacity is to be built. Among older people and women, the level of privacy-worry remained far lower than that of younger and male counterparts, with their digital literacy levels falling behind as well. It is important to note that digital literacy tested generic familiarities with basic terminologies. The large variance ($SD = 16.96$) and the low level of digital literacy ($M = 49.09$, out of 75) suggest some demographic groups may not have possessed even the basic understandings about overall digital ecosystems. That is, the baseline status of digital wellbeing upon which to build related preparedness or confidence in such matters as digital remains was found to be insufficient and even unequally distributed among the public.

The first set of hypotheses seeks to verify (1) the influences of demographic-socioeconomic backgrounds on access and (2) the influence of digital access on digital readiness. Supporting both hypotheses, the findings in Table 2 indicate the associations between digital access and general preparedness ($\beta = .07$, $p < .05$; H1a) and between

socioeconomic-demographics and access (H1b). In parceling the results for H1a, however, we found no support for privacy preparedness ($\beta = .03$, ns). In addition, in the Model 2 that included socio-demographics, we found no relationship between access and overall preparedness, indicating that access alone, varied by socio-demographics, might be a tenuous ground to build digital capacities.

The second hypothesis seeks to establish the indirect relationship between access and individual readiness via digital literacy. As shown in the last six columns of Table 2, we found supports. There was a positive role of access on digital literacy ($\beta = .24$, $p < .001$), which in turn produced better digital readiness. Digital literacy was the most consistent and strongest predictor in the models explaining both measures of digital readiness. The variable of digital literacy alone explained a large portion of the variance ($\beta = .20$ and $.29$, $p < .01$, with adjusted $R^2 = .04$ and $.08$), and the effect sizes did not vary with additional inclusions of access (Model 2) and socio-demographics (Model 3). In contrast, the change in the effect size of access was noteworthy, with the incremental addition of variables in the models for privacy preparedness ($\beta = -.10$, $p < .01$). That is, holding constant such factors as socio-demographics, the role of access turned negative, displaying the potential mediating power of digital literacy in pulling the influence of access into a different direction.

Table 3 displays the results concerning the third hypotheses, which seek to verify the indirect influence of digital literacy through digital readiness on perceived wellbeing (H3a) and their potentially compound patterns (H3b). We found strong supports for both. First, there were sizable functions of digital readiness ($\beta = .35$, general preparedness; $\beta = .34$, privacy preparedness, $p < .001$). The direct influence of digital literacy on perceived wellbeing was also positive ($\beta = .22$, $p < .001$). Importantly, however, in the full models that included all variables, we found divergent patterns of indirect influence of digital literacy through general and privacy preparedness: The effect of digital literacy on wellbeing was reduced substantially but remained significant with the inclusion of general preparedness ($\beta = .08$, $p < .05$). On the other hand, with the inclusion of privacy preparedness,

Table 2. Formation of Digital Wellbeing (DV): Access, Literacy, and Readiness.

	H1a access	H1b general prepared		H1b privacy prepared		H2 literacy	H2 general prepared			H2 privacy prepared		
		Model 1	Model 2	Model 1	Model 2		Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Female	.02		.02		.01	-.08**			.03			.03
Age	-.21***		-.04		-.16***	-.27***			.00			-.08*
Non-White	-.12***		-.04		-.07*	-.01			-.04			-.06 [^]
HH income	.18***		.10*		.12***	.08*			.08*			.10*
Education	.09*		.09*		.02	.14***			.07 [^]			-.01
Access		.07*	.02	.03	-.03	.24***	.00	-.02		-.07*		-.10**
Literacy							.20***	.20***	.17***	.29***	.32***	.28***
N	815	823	815	818	810	815	823	823	815	818	818	810
Adjusted R ²	.108	.005	.030	.000	.037	.223	.041	.040	.054	.086	.090	.100

[^] $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 3. Consequences on Non-Digital Perceived Wellbeing (DV).

	H3a-non-digital wellbeing		H3b-non-digital wellbeing		
	Model 1	Model 2	Model 1	Model 2.1	Model 2.2
Female		-.03		-.03	-.03
Age		-.05		-.03	-.00
Non-White		-.03		-.03	-.02
HH income		.18***		.17***	.17***
Education		.11**		.10**	.12**
Access		.03		.01	.03
Literacy			.22***	.08*	.05
Digital readiness					
General prepared	.35***	.30***		.29***	
N	823	815		815	
Adjusted R ²	.126	.204		.208	
Privacy prepared	.34***	.30***			.29***
N	818	810	823		810
Adjusted R ²	.117	.203	.049		.204

* $p < .05$, ** $p < .01$, *** $p < .001$.

the explanatory power of digital literacy became non-significant in the model. This contrasting finding illustrates the robust function of digital literacy in predicting privacy preparedness, also lending support for compound patterns of general and privacy preparedness in mediating the influence of digital literacy.

The fourth hypothesis addresses the interactions between digital readiness and (1) age and (2) privacy-worry. We found consistent and sizable interactions between age and two measures of readiness ($\beta = -.49$, $p < .001$, for overall preparedness; $\beta = -.39$, $p < .001$, for privacy preparedness). The interaction was also found to be significant between privacy-worry and overall preparedness ($\beta = .34$, $p < .001$), although the interaction with privacy preparedness ($\beta = .19$) was marginal at the level of .10. The patterns observed in Figure 2 reveal that unequal distribution of wellbeing exacerbates in certain non-digital (age) or digital (privacy-worry)

condition. As illustrated in the upper two panels, for older people, their wellbeing is more likely to suffer when they are not ready for either general or privacy preparedness. Similarly, for those with high privacy-worry, perceived wellbeing decreases with decreased overall preparedness, while magnifying the difference from people with low privacy-worry.

Finally, the results from SPSS PROCESS confirmed the hypothesized relationships in sequential and compound manners. We first ran PROCESS Model 6 to summarize the sequence of influences from digital access, digital literacy, and digital readiness to perceived wellbeing, keeping constant socio-demographics. Analyses, based on 10,000 bootstrapping samples with 95% biased corrected CIs, found no direct role of access, but its indirect influences were found—with general preparedness, $b = .020$ (.007), CI [.008, .037], and with privacy preparedness, $b = .034$ (.008),

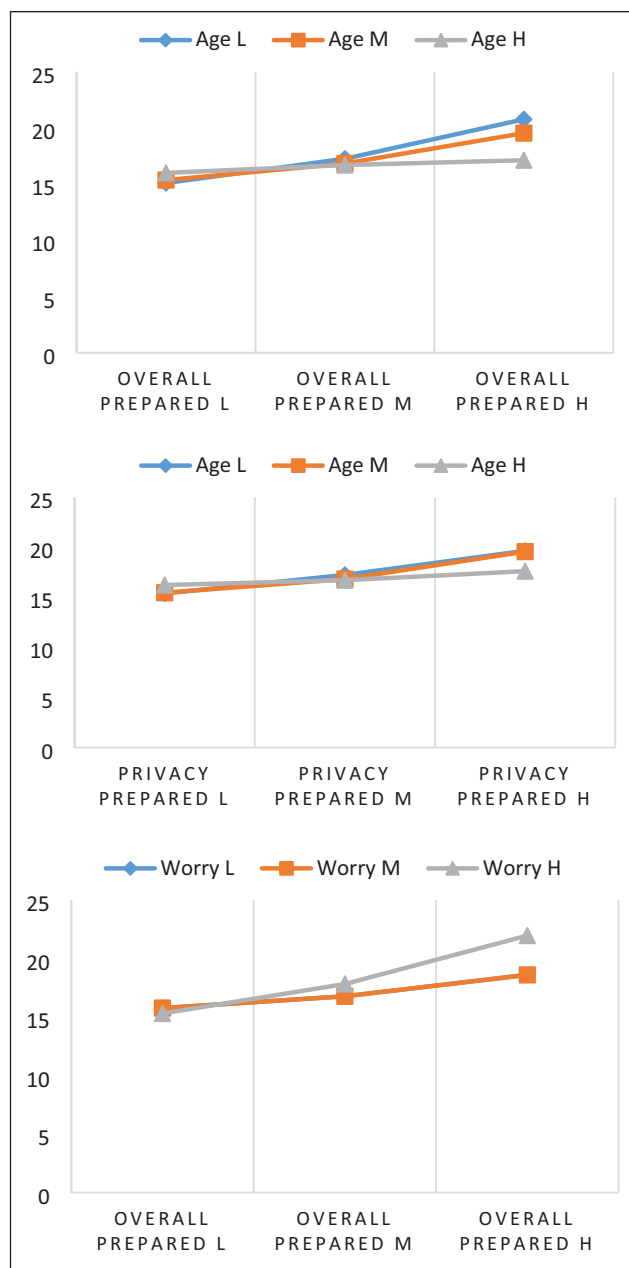


Figure 2. Interactions with age and privacy-worry.

Note. Plotted are mean scores broken down by distributions (low, middle, and high) of preparedness and age and privacy-worry.

CI [.019, .054]. In highlighting the compound links to well-being, PROCESS Model 14 tested four moderated mediations—the indirect roles of digital literacy via the readiness of general and privacy preparedness as their relationships are conditioned by the two moderators of age and privacy-worry. The results validated the findings of step-wise models. As found earlier, we found no significant mediation for privacy preparedness moderated by privacy-worry, but there was the indirect influence of digital literacy conditional upon age, $b = -.0003$ (.0001), CI [−.0006, −.0001], general preparedness; $b = -.0004$ (.0002), CI [−.0008, −.0001], privacy

preparedness. The indirect role of digital literacy was also conditional upon privacy-worry for general preparedness, $b = .0081$, (.0007), CI [.0005, .0034].

Discussion: Theoretical and Policy Implications

Evaluating the formation and consequences of digital after-life preparedness, we conceptualized the first-, second-, and third-level digital divide as individual capacity building in the process of developing one's wellbeing (Chen et al., 2014). In doing so, we also modified the notion of sequential and compound digital exclusion (van Deursen & Helsper, 2015) that underlies (1) the formation of digital wellbeing, as indicated by digital access, digital literacy, and digital readiness for afterlife, and (2) their consequences on non-digital wellbeing. That is to note, the different levels of digital divides are connected in highly sequential manners, producing the cumulative (dis)advantages in capacities among those who do (and do not) happen to possess particular digital resources. We related these sequential outcomes, interlaced in intermediate steps, to the life outcome of wellbeing—detecting the influence of one's digital status producing systematic differences and, thus, spilling over inequalities into other areas of life concern such as the afterlife digital remains (Evans, 2018). The theoretical distinction among the first-, second-, and third-level divides helps explain why there are widespread inequalities in acquiring cumulative advantages across a broad range of online/digital and non-digital life areas. It also clarifies the nature of interconnectedness from access, digital literacy, and types of individual capacities to tangible outcomes whereby socio-demographic contexts persistently reproduce inequalities.

To this end, this study hopefully represents a step toward understanding the consequences of unequal distribution of digital resources. In fact, we found that pre-existing inequalities reinforce the development of what one might call “digital repertoires”—the possession of distinct digital resources and capitals that are linked to incubate individual capacity building (Bourdieu, 1984; Helsper, 2012), as applied to the context of digital remains (Bassett, 2022; Birnhack & Morse, 2022; Harbinja, 2022; Öhman & Floridi, 2018). All hypothesized links turned out to be significant except that (1) the relationship between access and readiness for digital remains became tenuous, taking into account socio-demographics, and (2) there was no moderated mediation for privacy preparedness (digital literacy → readiness for digital remains, moderated by privacy-worry). This means that in preparing for our digital remains, in its ontology after death, we have realities in which many people are in fact not prepared. In other words, there are only few who are well positioned for dealing with or planning for digital remains, so long as they are digitally literate and already have higher digital access.

There can be several explanations for anomalous findings. First, when it comes to the tenuous relationship

between access and readiness, it rather indicates robust influences of socio-demographics on access as their explanatory power entirely took away the significance of access. Given that we found significant sequential indirect relationships via digital literacy, this also means that access to the basic technical basis may remain as the necessary (but not sufficient) first step to build readiness (thus, wellbeing). Second, in terms of non-significant moderated mediation, it may be that privacy-worry is too closely correlated with digital literacy to isolate effect. In fact, two variables turned out to be correlated and, thus, not strictly orthogonal ($r = .14, p < .001$)—that is, the empirical condition for interaction. Aside from privacy preparedness, however, we found all other interactions to be significant, demonstrating compound patterns as predicted. Thus, the distinction between compound dimensions of digital readiness should not be out of the equation explaining how one type of readiness might be different from other similar types in their respective influences.

Collectively, our findings point to the mechanism of perpetuating inequality: unequal access and unequal return. This is to note the presence of interdependent links from the entry point of digital access, unequally distributed, to the end point of non-digital wellbeing status, unequally returned. The void of the direct link from access to non-digital wellbeing ($\beta = .01, ns$) illustrates that key to the puzzle may be the intermediate steps in which digital literacy incubates individual readiness to cope with digital remains and, thus, to navigate digital ecosystems effectively, which in turn influences one's quality of life. Privacy paradox, as applied to digital post-life management, is important to recognize in this context. In fact, previous studies (Morse & Birnhack, 2022) found privacy paradox (high concern, but low action) was also found in the context of post-life management issues. The findings from this study in fact supported their work's findings, in that if there is any disjuncture between concern and management or planning, that is probably due to the lack of awareness or literacy. Harbinja (2022) even in a very practical level raised this concern about the difficulty of individuals actually managing the privacy of online identity, data protection, and access to data after death, suggesting the need to put together regulatory protection as much as practical digital tools for individual control.

This is not to downplay the importance of promulgating digital access and material connections at physical levels but to recognize the incremental (dis)advantages cumulated on top of basic access points, which enables users to translate them into further advantages. What is amazing about these findings is that even after controlling for socio-demographics, the sizable effects of digital literacy and digital readiness on the final outcome variable of wellbeing remain. This is in line with the empirical findings and conceptual advancement by the prior studies in the field of digital inequality that put an emphasis on skills, proficiency, and emergent forms of digital capitals (Helsper, 2012; Wendt et al., 2023). In this sense, the policy intervention to fix the unequal return of

access should be made beyond possible measures to eradicate socio-demographic divides and increase access.

Fundamentally, we argue that the capacity of one's preparedness for digital afterlife can represent the emergence of a third-level divide with its real-life consequences, or even a newer space for inequality debates in the future. Our concern is that previous policy discussions used to focus exclusively on the binary distinction between "haves" and "have-nots" at the infrastructural level. Such a narrow focus will not easily classify this new area of social concern (van Deursen & Helsper, 2015) such as the afterlife digital remains. This is the failure of not only policy but also societies in recognizing complex techno-social aspects of a person's afterlife (Kneese, 2023). That is to say, commercial platforms have always been part of experiences of death, as social media technologies exercise the power to influence not only over people's lives but also over their experiences of death (Bassett, 2022; Harju & Huhtamäki, 2023; Park, 2021). Thus, the attempt to address this issue solely as an individual issue for their own protection amounts to the failure of techno-solutionism—meaning that (1) individuals may be likely ill-prepared, and (2) commercial platforms themselves remain inadequate in providing technological means as well (Harju & Huhtamäki, 2023).

One of the recent studies' findings (Morse, 2024) indicated that the public still hesitates to adopt technologies for protecting their digital remains. This finding is important to address here because our finding also supports such findings that not every ordinary user is not be ready. Although our study's dependent variable was only concerned about the individual-level afterlife readiness for themselves, the divide found in our study indicates that even those who might be willing to take actions will be most likely in privileged social positions (Park et al., 2020).

As the rapid transition into algorithm-based digital participation continues, we will encounter this type of complex socio-policy issues continuously emergent on top of ubiquitous access. Furthermore, we expect that digital ecosystems intensify the demand for perpetual traces of personal data enabling deep looks into private lives, as illustrated in the previous works by Evans (2018), Morse (2018), and Morse & Birnhack (2022). Thus, it is troublesome to make no further policy effort to regulate digital platforms in their algorithmic use of information, which in its nature defies one's biological lifespan. This study informs the necessity of devising careful interventions that will improve on individual capacities to protect digital remains constituting one's digital identities even after death (Engler, 2022; Öhman & Floridi, 2018). The fact that only a relatively small number of people in highly privileged social positions were found to be ready, aware, or worried about the afterlife digital remains does not signal little significance of this issue. The lesson here is the apparent need to design effective interventions not just to make digital readiness accessible only for those elite users (Park, 2021b).

Normative policy objectives will be that the high status of digital or non-digital wellbeing should not be kept excluded

from those in less privileged social positions. Similarly, another point to take away from this study's findings is the persistently limited extent of digital literacy among those with marginalized backgrounds, notably of low levels of education and income. Past studies (e.g., Morse & Birnhack, 2022) identified the posthumous privacy paradox, the incongruence between behavior and concern applied to afterlife privacy issues. Privacy paradox will be the important issue to be studied continuously. Given the power of digital literacy (Hargittai & Hsieh, 2008) in relaying the benefits of digital access to other areas such as readiness for digital remains, future studies should delve into identifying the salient effects of knowledge in related areas beyond what is concerned in this study. In this vein, our explanation, although limited by available measures in capturing different forms of digital capitals of readiness and literacy, hints that the paradox of afterlife privacy protection may be also explained by digital literacy.

Ideally, the better model in the future will use a separate knowledge measure regarding digital remains to eliminate potentially compound dimensions that might conflate the functioning of digital literacy. Such detailed measures would allow future work to detect even-more-precise sequential patterns from digital literacy to individual readiness by looking at each disparate route (e.g., digital literacy → readiness in factual knowledge vs. readiness in self-perception → perceived wellbeing). It will be important to measure actual behavioral strategies to cope with digital remains, while other studies strengthen a different leg of this study's proposed model.

Finally, the limit of our data collection year (2018) should be noted. The core technologies of social media platforms have remained the same, but they also rapidly evolved. Future works will make valuable contributions by capturing and validating (1) people's use of newer technological features, such as legacy contact, and (2) overall algorithmic (afterlife) readiness of which the measure might encompass other capacities in response to increasingly machine-learning-driven environments. Overall, our findings open the door to further investigations into related areas of specific data concerns previously unanticipated in non-algorithmic environment. For the concern under the current study, our evidence sheds a critical light on the newer digital spaces where the limits of humans as a biologically finite being will pose a disjuncture between individual capacity to control (thus, create wellbeing) and the data-based algorithm defiantly carrying our identities, traces, and privacy.

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